Name _____ Period _____ Date _____

3-4 Lesson Quiz: Dividing Polynomials

short form

2. Morgan begins to use synthetic division to divide $x^4 + x^3 - 6x^2 - 4x + 8$ by $x-2$, as shown at the right. Which of the following shows the correct values to replace the question marks, and also gives the quotient?

Ⓐ 2, 6, 0, −8; $x^3 - 3x^2 + 4$

Ⓑ 2, 6, 0, −8; $x^3 + 3x^2 - 4$

Ⓒ 1, 3, 0, −4; $x^2 + 3x - 4$

Ⓓ 1, 1, −6, −4; $x^3 + 3x^2 - 4$

2	1	1	−6	−4	8
		?	?	?	?

Show your work: finish the synthetic division in the table.

3. Verify the Remainder Theorem if $P(x) = 2x^3 - x^2 + 4x + 5$ is divided by $x + 2$. Complete the sentences.

When $P(x)$ is divided by $x + 2$, the remainder is
◯ −23 ◯ −7 ◯ 1 ◯ 33

$P(-2) =$ ◯ −25 ◯ −23 ◯ 9 ◯ 25

Does this example verify the Remainder Theorem?
◯ Yes ◯ No

Show your work (synthetic division and $P(-2)$):

5. If $x + 2$ is a factor of $P(x) = 2x^3 + 5x^2 + 5x + 6$, select the two factors that express $P(x)$ as a product.

☐ **A.** $(x + 2)$

☐ **B.** $(2x^2 - x - 3)$

☐ **C.** $(2x^2 - x + 3)$

☐ **D.** $(2x^2 + x - 3)$

☐ **E.** $(2x^2 + x + 3)$

☐ **F.** not a factor

Show your work:

How does your work show that $x+2$ is a factor?

Scoring: Question 2 and Question 5 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20. (Question numbers are the Savvas Lesson Quiz numbers; items 1 and 4 are not on this form.)

3-4 Lesson Quiz, short form: answer key

The question numbers are the Savvas Lesson Quiz item numbers, as on the packet's "On the quiz" box and IXL table. Items 1 and 4 are left off (below).

2. B: 2, 6, 0, −8**;** $x^3 + 3x^2 - 4$**.** Bring down 1; $2 \cdot 1 = 2$, $1 + 2 = 3$; $2 \cdot 3 = 6$, $-6 + 6 = 0$; $2 \cdot 0 = 0$, $-4 + 0 = -4$; $2 \cdot (-4) = -8$, $8 + (-8) = 0$. Bottom row 1, 3, 0, −4 | 0, so the quotient is $x^3 + 3x^2 + 0x - 4$ with remainder 0. A flips the signs of the last two quotient terms, C puts the bottom row where the products go and drops a degree, D copies the top row. Savvas quiz item 2, DOK 2, Realize skill A75. Taught by Example 2. **10 points:** 7 for B, 3 for the completed table.

3. −23**;** −23**; Yes.** Synthetic division by $x - (-2)$: bottom row 2, −5, 14 | −23. Direct substitution: $P(-2) = 2(-8) - 4 + 4(-2) + 5 = -16 - 4 - 8 + 5 = -23$. The remainder equals $P(-2)$, which verifies the Remainder Theorem. Distractor 25 in the second sentence is $P(2)$, the sign error with $x + 2$. Savvas quiz item 3, DOK 2, Realize skill A74. Taught by Example 5 (Methods 1 and 2 give the same number; Example 3 is cut). **Bonus 5:** all three correct 5; two correct 3; otherwise 0.

5. A and E: $(x + 2)(2x^2 + x + 3)$**.** Synthetic division with $a = -2$: bottom row 2, 1, 3 | 0. The remainder is 0, so $x + 2$ is a factor (Factor Theorem) and the quotient is $2x^2 + x + 3$.
Check: $(x + 2)(2x^2 + x + 3) = 2x^3 + 5x^2 + 5x + 6$. Savvas quiz item 5, DOK 2, Realize skill A75. Taught by Example 5A. **10 points:** 4 for each correct factor and none wrong, 2 for the work showing a remainder of 0.

Items left off this form (available for a retake or Form B): Savvas item 1, use long division to divide $x^3 + x^2 - 2x + 14$ by $x + 3$, build the answer from the choices x^2, $2x$, $-2x$, $4x$, −8, 4, 10, $-\frac{10}{x+3}$, $\frac{2}{x+3}$, $\frac{38}{x+3}$, $\frac{44}{x+3}$ (answer $x^2 - 2x + 4 + \frac{2}{x+3}$; DOK 2, skill A74; Example 1 long division, cut from the packet, though synthetic division with $a = -3$ gives the same result); Savvas item 4, the remainder when $f(x) = x^4 - 5x^2 - 6x - 10$ is divided by $x - 3$, choices A −46, B 8, C 44, D 224 (answer B, 8; DOK 2, skill F01; Example 4, rehearsed by Example 5 and IXL CHC).

IXL replacement: SmartScore 70 or above on D6D (item 2), CHC (bonus item 3) and VXE (item 5) replaces this quiz with 20/20.